RUDRA CHAVDA

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EDUCATION

University of California, Santa Barbara — Expected Spring 2028
Bachelor of Science, Computer Engineering — **GPA: 3.86**

University of California, Santa Cruz — 2024 – 2026
Bachelor of Science, Computer Engineering — **Honors:** 6x Dean's List

Coursework: Systems programming, computer architecture, operating systems, data structures and algorithms, and software engineering.

TECHNICAL SKILLS

Languages:	Python, C, C++, Java, TypeScript, JavaScript, SQL, RV Assembly
Frameworks:	Next.js, React, Vue.js, Spring Boot, Node.js, Py4Web, Tailwind CSS, Angular
Tools:	Git, Docker, Kubernetes, MySQL, Flyway, Microsoft Graph API, JUnit, CARLA, GDAL
Concepts:	LLM Agents & Multiagent workflows, Function Calling, OAuth 2.0, Reinforcement Learning, System Design, Data Structures, RAG, MCP Servers, Prompting, Guardrails and prompt injection, Quantization and distillation.

RELEVANT EXPERIENCE

Ascend Cargo — Jun 2026 – Sep 2026
AI/ML Software Engineering Intern — *Remote, San Francisco Bay Area*

- Architected an enterprise **LLM agent framework** utilizing **Java 23, Spring Boot 3.4, and OpenAI function calling**, featuring annotation-driven tool registration and dynamic JSON-Schema generation for targeted intent routing.
- Engineered a natural-language shipment management system, diagnosing and resolving a synchronous automation bottleneck that reduced database execution latency from **12 minutes to under 30 seconds**.
- Designed a Role-Based Access Control (RBAC) authorization pipeline enforcing strict per-user AI limits (**NONE/READ/WRITE**) and integrated a **Microsoft Graph API** inbound email channel via **OAuth 2.0** for authenticated, remote **shipment creation and updates**.
- Hardened agent fault tolerance by implementing explicit refuse-and-instruct error contracts, and authored **122 JUnit/Mockito unit tests** alongside a custom CLI harness to validate state mutations against the database.

AIEA Lab (UCSC) — Dec 2025 – Apr 2026
Undergraduate Researcher — *Onsite, Santa Cruz*

- Benchmarked and wrote Reinforcement Learning (RL) algorithms including **DQN, DDQN, A3C, and PPO** to evaluate autonomous vehicle safety parameters within the Robustifying AVs project.
- Deployed and scaled the **CARLA** autonomous driving simulator in a headless environment across the **Nautilus Kubernetes** hyper-cluster, isolating edge-case failure modes via post-training reward function visualization.

CA-Futures (UCSC EEB & Baskin Engineering) — May 2025 – Sep 2025
Software Developer — *Hybrid, Santa Cruz*

- Engineered a full-stack climate modeling platform utilizing **Vue.js** and **Py4Web**, delivering interactive web maps that enable researchers to visualize and compare predictive ecological shifts across multiple long-term climate scenarios.
- Architected an asynchronous geospatial pipeline utilizing **GDAL, Rasterio, and Geopandas** to intercept user-drawn map polygons, compute dynamic habitat connectivity, and implement output caching to eliminate redundant backend execution.

Opennote (YC S25) — Jul 2025 – Feb 2026
Campus Growth Lead — *Onsite, Santa Cruz*

- Spearheaded campus integration strategies, driving a **30% increase** in user engagement and securing **50+ daily active users** by organizing targeted technical workshops on TypeScript user-agent API flows.
- Co-directed the UCSC Cruzhacks integration, managing vendor partnerships and allocating **$1,500+** in technical resources to drive platform adoption among computer science students.

Fassur — June 2023 – Oct 2023
Front-End Engineering Intern — *Remote, SF Bay Area*

- Engineered a scalable, responsive web application utilizing **React.js** and **Tailwind CSS**, developing a modular component library that accelerated frontend feature deployment.
- Architected interactive client dashboards dedicated to complex **data visualization**, optimizing DOM rendering logic to process high-volume datasets while maintaining strict UI responsiveness.
- Integrated **RESTful APIs** with robust frontend state management to ensure seamless data synchronization, reducing redundant network overhead and improving initial page load speeds.